Open Science and Data Management Plan

1. Data and software the project will use

All input data are free, public, and archived by their originating programs; nothing is purchased or restricted.

Input	Archive / host	Access
2001 airborne lidar DEM (NASA ATM, 2 m)	USGS ScienceBase	public
2014–15 airborne lidar DEM + point cloud (NCALM, 1 m)	OpenTopography	public
REMA satellite DEM v2 (2 m mosaic, 2021–23)	Polar Geospatial Center / AWS Open Data	public
ICESat-2 laser altimetry (ATLAS; ATL06/ATL08), external geodetic control	NASA NSIDC DAAC	public
Permafrost / ground-ice and microclimate-zone maps (substrate covariates)	Published MDV mapping (Bockheim et al. 2007; Fountain et al. 2014)	public
Stream discharge (21 gauges), meteorology, glacier mass balance, soil climate	MCM-LTER via the Environmental Data Initiative (EDI)	public
Stream-channel outlines (labels)	EDI (knb-lter-mcm scope)	public
ERA5 reanalysis	Copernicus Climate Data Store	public
AMPS Antarctic mesoscale model output	NCAR GDEX	public

One dataset in this domain is *not* public: the prior dissertation's hand-digitized training tiles, held by their author. If shared, they will be used under the author's terms and will not be redistributed, and all credit will remain with the author.

All processing software is open source: Python (numpy/scipy, rasterio, geopandas, PyTorch), PDAL, GDAL/OGR, and WhiteboxTools. No proprietary software is required to reproduce any result.

2. Data and software the project will produce

Product	Format / standard	Where released
Elevation-change (DoD) rasters, all epoch pairs	Cloud-Optimized GeoTIFF, explicit CRS (EPSG:3294), nodata and units in metadata	Zenodo
Per-stream erosion/deposition rate tables	CSV with data dictionary	Zenodo
Per-pixel calibrated detection-threshold layers (Objective 3)	Cloud-Optimized GeoTIFF	Zenodo
Stream-channel outlines + process-classified geomorphic-change maps (Objective 2)	GeoPackage (OGC) / Cloud-Optimized GeoTIFF	Zenodo
Training labels (if self-digitized)	GeoPackage + tile rasters	Zenodo
Trained model weights	PyTorch checkpoint + ONNX export, with training configuration	Zenodo
Processing code (full pipeline: fetch → derivatives → detection → change → attribution)	Python, version-controlled public repository, OSI-approved license (Apache-2.0)	Public repository, archived to Zenodo

Driver-attribution model outputs (Objective 1)	CSV + fitted-model objects with documentation	Zenodo

Documentation standard. Every released raster/vector product carries: coordinate reference system (EPSG code), acquisition epochs and processing date, method summary, uncertainty estimate (NMAD/LOD95 or the Objective 3 per-pixel layer), and a pointer to the exact code version (commit/DOI) that produced it.

3. When and where products are shared

- Code: developed in the open in a public repository from award start; released versions archived with DOIs.
- Data products: released no later than the publication of the paper they support, and at yearly milestones regardless of publication status.
- Publications: submitted to open-access venues or archived as accepted manuscripts in a public repository, consistent with SMD open-access policy.

4. Licenses

Data products: CC0 or CC-BY. Software: Apache-2.0. Third-party inputs remain under their original terms and are cited, not re-hosted, except where their license permits convenience copies.

5. Protections and exceptions

The project uses no human-subjects data, no export-controlled data, and no commercially restricted data. The single access-limited item (the author-held label tiles) is handled as described in §1, namely use with permission, no redistribution, and a published replacement set if the project digitizes its own.

6. Roles and oversight

The FI is responsible for day-to-day data management, repository hygiene, and product releases; the PI reviews each public release. Working data are stored on institutionally backed storage with an off-machine mirror; the public archives provide long-term preservation beyond the award period.

7. Reproducibility commitment

Every number and figure in project publications will regenerate from public inputs via the released code: a single documented command sequence per product. Heavy intermediate rasters are treated as regenerable and are not archived; the scripts and small tables/figures that define them are.